25900336 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSECFS201 - OPERATING SYSTEMS (SPECIALIZATION : OPERATING SYSTEM ARCHITECTURE)

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Explain batch operating System with an example. [U] / [CO1]
2. Examine mutual exclusion. [An] / [CO1]
3. Analyse throughput to overall CPU scheduling efficiency. [An] / [CO1]
4. Define segmentation in memory management. [U] / [CO2]
5. List any two page replacement algorithms. [K] / [CO2]
6. Illustrate how paging within each segment reduces external fragmentation. [U] / [CO2]
7. Distinguish between inode and directory entry. [An] / [CO3]
8. Analyse how tree-structured directories reduce name conflicts compared to two-level directories. [An] / [CO3]
9. Examine the relationship between spooling and concurrent process execution. [An] / [CO3]
10. Explain the importance of text editors in Linux. [U] / [CO4]

11. Explain hard link and soft link in Linux. [U] / [CO4]

12. Restate the vi command for deleting the current word. [U] / [CO4]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

13. Describe the lifecycle of a thread, including its various states. [U] / [CO1]

14. Explain how the bootstrapping process works. [An] / [CO1]

15. What is meant by inter process communication? [U] / [CO1]

16. State the differences between safe and unsafe state in deadlock. [An] / [CO2]

17. Explain how paging works? [An] / [CO2]

18. Explain various strategies for handling deadlock [An] / [CO2]

19. Analyse the role of the kernel I/O subsystem in coordinating device drivers and user processes. [An] / [CO3]

20. Explain the hardware requirements for Linux OS. [An] / [CO4]

[5x6 = 30]

Part C

Essay Type Questions

Answer any **2** questions

Each question carries **10** marks

21. Write a brief description of First Come First Serve CPU Scheduling Algorithm. [U] / [CO1]

22. Apply the concept of virtual memory to explain how a 32-bit logical address space can be mapped onto a smaller physical memory using paging. [An] / [CO2]

23. Differentiate between programmed I/O, interrupt-driven I/O, and DMA, and evaluate which is most efficient in high-speed data transfer. [An] / [CO3]

[10x2 = 20]